

Researchers in Michigan Aim to Grow Stronger Apple Trees

Researchers in the U.S. state of Michigan are looking for ways to help apple trees survive bad weather. To do this, one researcher plans to use a late bloomer.

The late blooming tree is a native Michigan apple tree known by the scientific name *Malus coronaria*. To fight freezing temperatures, the tree blooms two or three weeks later than trees that produce cultivated varieties of apples like Honeycrisp or Red Delicious.

“That doesn’t sound like a lot, but almost always that’s enough for the flowers to escape the killing spring frost,” said Steve van Nocker. He is a professor and plant geneticist at Michigan State University.

Van Nocker wants to identify the genes responsible for the *Malus coronaria*’s delayed blooms. Then he will use those genes to develop more frost-resistant apple varieties.

Van Nocker’s project is one of many ways researchers and growers are trying to make apples more resilient. They say this is important because of climate change.

Other researchers are making similar efforts.

Jared Buono is the director of Cornell University’s Hudson Valley Research Laboratory in New York state. Buono’s lab is adding solar panels to its experimental orchard. He said growers are experimenting with solar panels in Europe and South Korea. The panels protect fruit from hail and sun damage but also let in light and heat when necessary.

University of Maryland researchers recently announced the development of a more heat-tolerant apple variety. At Penn State University in Pennsylvania, a team built an unmanned vehicle that uses propane fuel to warm trees up as it moves around an orchard.

Buono said these efforts can ease the effects of bad weather. On May 18, 2023, frost damaged trees across New York state and cut apple production by 20 percent.

A recent Washington State University study examined weather in apple growing areas from 1979 to 2022. It said the top apple producing states of Washington, Michigan and New York reported fewer cold days and more warm fall nights than they used to. Washington state also has reported more extreme heat.

The study said warming temperatures could cause trees to come out of dormancy too early and increase the risk of sunburn. This hurts the quality of apples.

Malus coronaria is also called sweet crabapple. It was once so common in Michigan that its blossoms were named the state flower in 1897. But, van Nocker said it appears to be disappearing. The trees that remain often produce little new growth or fruit.

Van Nocker and graduate student Lily Christian recently took cuttings from some trees. They will test them to make sure they are fully *Malus coronaria* and not a hybrid variety. Then, the cuttings will be grafted onto apple rootstock. They hope to get healthy trees using this method.

Their goal is to identify the genes that cause late blooming and create a variety of apple that can be sold in markets. Van Nocker is also sending seeds to the U.S. Department of Agriculture. If the tree continues to disappear, he hopes to save its genes this way.

Making a later-blooming commercial apple could take more than 10 years. If a *Malus coronaria* is crossbred with a commercial apple tree to make a new hybrid, it would take four or five years for that tree to start producing apples.

Crossbreeding would continue until researchers developed an apple with the desired qualities of both varieties.

But van Nocker said it is important to try.

“With climate change, it’s quite possible that apples won’t be able to be produced in Michigan in the future,” he said. “That’s why cultivar improvement, especially in terms of frost resistance, is so important.”

I’m Anna Matteo.

Dee-Ann Durbin reported this story for the Associated Press from Washington. Anna Matteo adapted it for VOA Learning English.

Vocabulary

bloomer - n. พืชที่ออกดอกในช่วงเวลาหนึ่งของปี

cultivated variety - n. พืชที่ได้รับการพัฒนาเป็นพิเศษเพื่อใช้ในการเกษตรหรือในสวนของบุคคล

frost - n. ชื้นน้ำแข็งที่ก่อตัวบนพืชเมื่ออุณหภูมิต่ำกว่าจุดเยือกแข็ง ซึ่งสร้างความเสียหายต่อผลไม้

resilient - adj. สามารถทนต่อแรงกระแทกหรือการปฏิบัติที่ไม่ดี และกลับคืนสู่สภาพปกติได้

orchard - n. พื้นที่หรือฟาร์มที่ปลูกไม้ผล

solar panel - n. อุปกรณ์ที่เปลี่ยนแสงอาทิตย์ให้เป็นกระแสไฟฟ้าตรง

tolerant - adj. สามารถทนทานต่อสภาพที่ยากลำบากหรือเลวร้ายได้

dormancy - n. สภาพที่พืชไม่เจริญเติบโตในช่วงฤดูหนาวหรือสภาพแห้งแล้ง แต่เก็บพลังงานไว้สำหรับฤดูเพาะปลูก

hybrid - n. พืชหลากหลายชนิดที่เกิดจากการรวมกันของสารพันธุกรรมจากพืชสองชนิดขึ้นไป

graft - v. การนำกิ่งตอนจากพืชหนึ่งไปติดกับอีกพืชหนึ่งเพื่อให้เจริญเติบโต ออกดอก และออกผล

rootstock - n. (พืชสวน) รากที่มีชีวิตที่รวมเข้ากับกิ่งของพืชอื่นที่คล้ายกัน ซึ่งจะเติบโตพร้อมกับคุณสมบัติของพืชทั้งสอง

crossbreed - v. การใช้วิธีการอย่างใดอย่างหนึ่งเพื่อรวมคุณสมบัติทางพันธุกรรมเพื่อสร้างพืชที่ได้รับการปรับปรุง

commercial - adj. ออกแบบมาเพื่อผลกำไรเป็นหลัก

Reading Comprehension Quiz

1. What is the primary focus of researchers in Michigan regarding apple trees?

- A) Developing new apple varieties for taste.
- B) Finding ways to help trees survive adverse weather.
- C) Improving the marketing of local apple produce.
- D) Studying the history of apple cultivation in the state.

2. What is a key characteristic of the *Malus coronaria* apple tree that helps it combat freezing temperatures?

- A) It produces fruit that is naturally resistant to frost damage.
- B) It blooms significantly earlier than commercial apple varieties.
- C) Its blossoms appear two to three weeks later than cultivated varieties.
- D) It has a thicker bark that insulates against cold.

3. According to Steve van Nocker, why is a two-to-three-week delay in blooming significant?

- A) It allows for a longer harvest season.
- B) It typically provides enough time for flowers to avoid destructive spring frosts.
- C) It ensures better pollination rates.
- D) It results in larger, sweeter apples.

4. What is Steve van Nocker's ultimate goal for using the genes from *Malus coronaria*?

- A) To transplant entire *Malus coronaria* trees into commercial orchards.
- B) To develop new apple varieties that are more resistant to frost.
- C) To genetically modify existing apple trees to bloom earlier.
- D) To study the historical spread of *Malus coronaria* across Michigan.

5. The efforts by Van Nocker and other researchers to make apples more resilient are primarily driven by what global concern?

- A) A decline in consumer demand for apples.
- B) The increasing cost of agricultural labor.
- C) The effects of climate change.
- D) A shortage of suitable land for orchards.

6. How are solar panels being used in experimental orchards, according to Jared Buono?

- A) To generate electricity for orchard equipment.
- B) To provide shade and protection from hail and sun damage.
- C) To warm the trees during cold periods.
- D) To enhance the growth of fruit through artificial light.

7. Which of the following is NOT an example of efforts made by universities to protect apple trees mentioned in the text?

- A) Development of a more heat-tolerant apple variety by University of Maryland.
- B) Use of an unmanned vehicle fueled by propane to warm trees at Penn State University.
- C) Introduction of new irrigation systems by Cornell University.
- D) Experimentation with solar panels at Cornell University's Hudson Valley Research Laboratory.

8. What impact did a frost on May 18, 2023, have on apple production in New York state?

- A) It caused a minor delay in harvest.
- B) It increased apple production by 20 percent.
- C) It resulted in a 20 percent cut in apple production.
- D) It primarily affected the quality, not the quantity, of apples.

9. What trend did a Washington State University study (1979-2022) identify in top apple producing states like Washington, Michigan, and New York?

- A) An increase in overall annual rainfall.
- B) Fewer cold days and more warm fall nights.
- C) A decrease in extreme summer heat.
- D) A shift in apple varieties being cultivated.

10. According to the study, how do warming temperatures negatively affect apple trees and their fruit quality?

- A) They cause trees to overproduce, leading to smaller fruit.
- B) They can lead to early dormancy exit and an increased risk of sunburn.
- C) They promote the growth of harmful pests and diseases.

D) They require more water, stressing irrigation systems.

11. What is the current status of *Malus coronaria* (sweet crabapple) in Michigan, despite its historical significance?

A) It is thriving and expanding its natural habitat.

B) It is being widely cultivated for commercial purposes.

C) It appears to be disappearing, with remaining trees producing little growth or fruit.

D) It has been replaced by more popular hybrid varieties.

12. What is the immediate step van Nocker and Lily Christian are taking with cuttings from *Malus coronaria* trees?

A) Planting them directly into new orchards to observe their growth.

B) Crossbreeding them with commercial apple trees immediately.

C) Testing them to confirm they are pure *Malus coronaria* before grafting them onto rootstock.

D) Analyzing their genetic code to identify specific resistance genes.

13. Besides creating a marketable apple variety, what other important action is van Nocker taking to preserve *Malus coronaria*?

A) Establishing new protected reserves for the trees.

B) Distributing the trees to home gardeners for wider cultivation.

C) Sending seeds to the U.S. Department of Agriculture to save its genes.

D) Developing a public awareness campaign about its decline.

14. Approximately how long would it take for a new hybrid tree, created by crossbreeding *Malus coronaria* with a commercial apple tree, to start producing apples?

A) Less than one year.

B) Two to three years.

C) Four or five years.

D) More than ten years.

15. What is the ultimate concern expressed by van Nocker regarding apple production in Michigan due to climate change?

A) A shift in consumer preference to other fruits.

B) The possibility that apples may no longer be produced in Michigan.

C) The need for increased government subsidies for apple growers.

D) The requirement to import all apples from other regions.